The Closed SIR Flow for All Positive Parameters: Phase Curves, Final-Size Branches, and the No-Recurrence Boundary

Route-A structural certificate C198

Abstract

We give one complete dynamical atlas for the closed mass-action SIR system at all positive transmission and removal rates. Exact scaling reduces the family to one planar flow; its first integral yields the phase curves, infection peak, time quadrature, final-size sensitivity and the correctly selected Lambert branch. A monotone removed coordinate proves global convergence and excludes every nonconstant recurrent orbit. A Lambert-free high-precision checker independently reconstructs both real final-size intersections. The result is an idealized mathematical certificate, with no clinical data or medical advice. Its monotonicity and absence of intrinsic prime arithmetic force strict Route-A rejection.

Keywords: SIR model; phase portrait; final size; Lambert W; threshold; monotone dynamics; Route-A obstruction.

中文摘要

本文对全部正传播率与移除率的封闭质量作用SIR系统给出统一动力学图谱。 精确无量纲化把全参数族化为一个平面流；第一积分同时确定相曲线、感染峰值、时间积分、 最终规模灵敏度及正确的Lambert W分支。移除变量严格单调，因而全局收敛并 排除全部非常值回归轨道。本文只研究理想化数学模型，不使用临床数据，也不提供医疗建议。

关键词：SIR模型；相图；最终规模；阈值；单调动力学。

1 One phase portrait for all parameters

Fix $\beta, \gamma > 0$ and physical time t in

$$\dot S = -\beta S I, \qquad \dot I = \beta S I - \gamma I, \qquad \dot R = \gamma I.$$

The model and its threshold/final-size analysis descend from Kermack and McKendrick [1]; Lambert-branch analysis is treated explicitly by Pakes [2]. Put

$$\kappa = \gamma/\beta, \qquad x = S/\kappa, \quad y = I/\kappa, \quad z = R/\kappa, \quad \tau = \gamma t.$$

Every parameter pair is thereby conjugate, without changing physical-time orientation, to

$$x' = -xy, \qquad y' = y(x-1), \qquad z' = y. \tag{1}$$

The nonnegative fixed-population simplex is invariant and compact, so solutions are global. Direct differentiation gives

$$H(x,y) = x + y - \log x = H(x_0, y_0), \qquad y = y_0 + x_0 - x + \log(x/x_0). \tag{2}$$

For $y_0 > 0$, x decreases strictly. If $x_0 > 1$, then y grows until the unique crossing $x = 1$ and

$$y_{\max} = y_0 + x_0 - 1 - \log x_0.$$

If $x_0 < 1$, y decreases from the start; at $x_0 = 1$ its first derivative is zero, after which decreasing x makes it negative. Thus the threshold and peak are globally, not locally, classified.

2 Global limit, branch, and stability

Since $z' = y \geq 0$ and population is bounded, $\int_0^\infty y\, d\tau < \infty$; bounded derivatives give $y(\tau) \to 0$. Also $x(\tau) = x_0 \exp(-\int_0^\tau y\, ds) > 0$. Hence every $y_0 > 0$ trajectory has $0 < x_\infty < \min(x_0, 1)$ and (2) gives

$$x_\infty e^{-x_\infty} = x_0 e^{-x_0 - y_0}, \qquad x_\infty = -W_0(-x_0 e^{-x_0 - y_0}). \tag{3}$$

The function $x - \log x$ decreases on $(0,1)$ and increases on $(1,\infty)$. Thus $-W_{-1}$ is the other, upper intersection of the invariant curve; it is not the forward limit. The exact time representation is

$$\tau = \int_{x(\tau)}^{x_0} \frac{du}{u\{y_0 + x_0 - u + \log(u/x_0)\}}.$$

Implicit differentiation of (3) yields $\partial x_\infty / \partial y_0 = x_\infty/(x_\infty - 1) < 0$.

If $y_0 = 0$, the initial point is already an equilibrium and remains fixed. In particular, when $x_0 > 1$ its final value is the upper root x_0, not the positive-infection W_0 branch. On each population simplex the equilibrium line has tangential eigenvalue zero and transverse physical eigenvalue $\beta S_* - \gamma$. Finally, R is strictly increasing wherever $I > 0$; there is no nonconstant periodic or recurrent orbit, and every trajectory converges to the disease-free line.

Revision focus. Round 1 adds global convergence, both Lambert branches, quadrature, sensitivity, equilibrium stability and no recurrence.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany this manuscript; no clinical data are used.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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References

[1] W. O. Kermack and A. G. McKendrick, "A contribution to the mathematical theory of epidemics," *Proc. R. Soc. A* 115 (1927), 700–721. DOI: 10.1098/rspa.1927.0118.

[2] A. G. Pakes, "Lambert's W meets Kermack–McKendrick epidemics," *IMA J. Appl. Math.* 80 (2015), 1368–1386. DOI: 10.1093/imamat/hxu057.